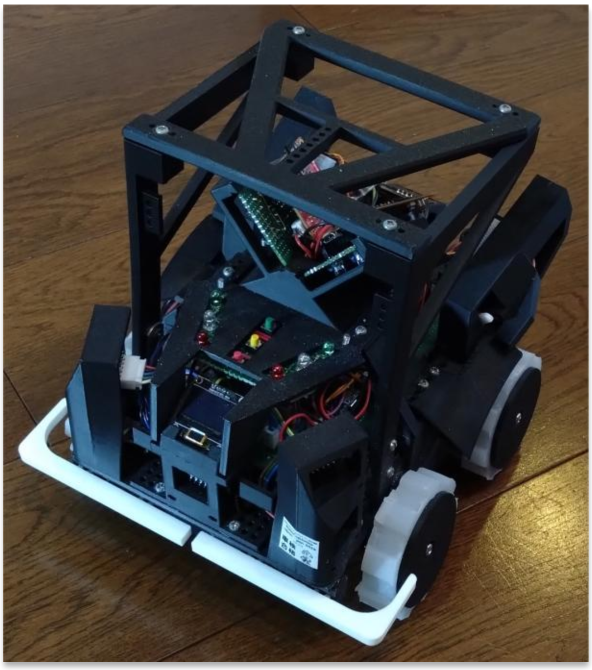
RoboCupJunior Rescue (Maze) 2026

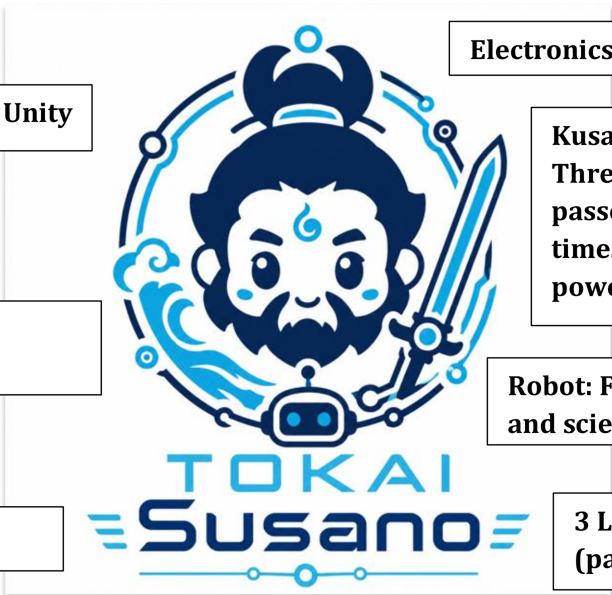
Team Description Paper

TOKAI Susano

Divine Eyes and Feet Guiding the Maze - Outstanding Traversability and Precision -



Team Members: Riku Harada / Yoshiharu Tsujimoto / Seigo Sobue



Abstract

TOKAI Susano is a team taking on RoboCupJunior Rescue Maze for the first time. Our initial goal was to place highly in the Japan Open, and through three prototypes we gradually developed stable driving, reliable exploration, victim detection, and rescue-kit deployment. The current robot combines a compact, low-center-of-gravity four-wheel-drive structure, self-made silicone tires, multiple ToF sensors, an IMU, a color sensor, and OpenMV cameras. In particular, we emphasized backup systems for dealing with errors and restarts during the competition, including tile-by-tile coordinate management, recording wall information, saving and restoring data with a microSD card, and coordinate matching after leaving the red zone. As a result, the robot has achieved a high return success rate in flat mazes and continues to be improved so that it can handle obstacles and multi-floor courses. This TDP describes TOKAI Susano's mechanical design, electronic design, software architecture, performance evaluation, and the problems encountered during development and how they were solved.

1. Introduction

a. Team

TOKAI Susano consists of three members from Tokai Junior and Senior High School. The team name Susano comes from Susanoo-no-Mikoto, a hero in Japanese mythology. Inspired by the story of Susanoo defeating the Yamata-no-Orochi and saving people, our goal is to bravely explore the maze, reliably find victims, and rescue them.

Name	Grade	Main Role	Experience / Contributions
Riku Harada	Grade 11	Leader / Integration / Strategy	FLL 2025 Japan Open Champion. Responsible for overall direction, driving strategy, integration testing, and development records.
Yoshiharu Tsujimoto	Grade 10	Hardware	FLL 2025 Japan Open Champion. Responsible for tires, body structure, assembly, and improving maintainability.
Seigo Sobue	Grade 9	Software	WRO 2024 Middle Category Japan Open Runner-up. Responsible for the AI camera, victim detection, and control programs.

In addition to practice at school and other facilities, the team prepared meeting materials at home and used Discord web meetings to advance robot development every day. This made it possible to develop the robot in a short period of time.

Meeting Materials

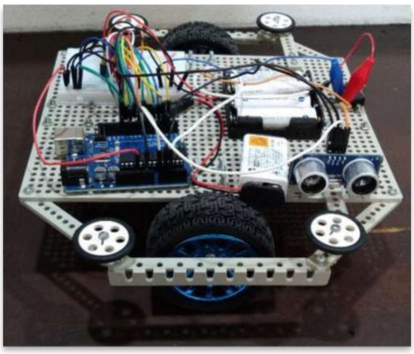
2. Project Planning

a. Overall Project Plan

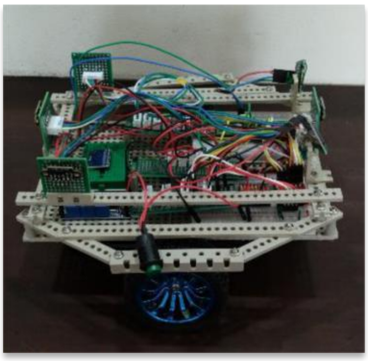
To build a robot capable of reliable exploration and return in a flat maze without obstacles, we built three prototype robots.

Based on the competition challenges and the development period, our team defined the following requirements.

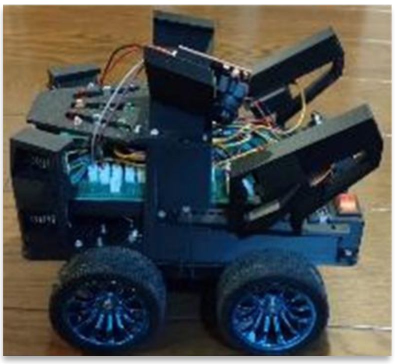
Category	Requirement	Reason
Driving Performance	Move stably in 30 cm units and handle walls and obstacles.	In Maze, the robot must move reliably from tile to tile in order to expand the exploration area.
Return Performance	Return to the starting point with a high probability after exploration.	To earn the exit bonus and improve reliability.
Coordinate Management	Restore coordinates after driving errors or after a restart.	Because coordinate deviation can occur during long runs and in the red zone.
Victim Detection	Detect letter victims and cognitive targets.	To score points for victim detection and rescue-kit deployment.
Maintainability	Use a structure that can be repaired and adjusted quickly.	To cope with the limited adjustment time during the competition.



Prototype 1



Prototype 2



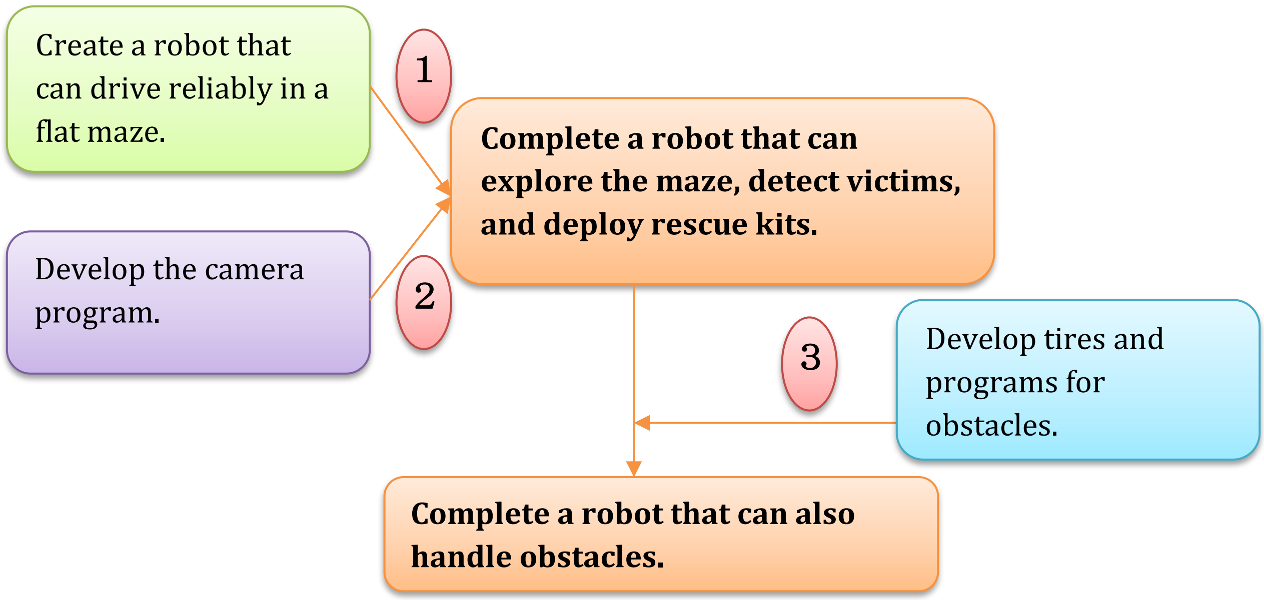
Prototype 3

Robot Development Milestones

Period	Major Milestone	Review Gate
July	Rule analysis, strategy and planning, Prototype 1 development, and construction of a 3 x 4 simple course.	Complete one lap of the course.
August	Prototype 2 development, introduction of ToF and color sensors, and construction of a 4 x 6 course.	Reach all tiles, including floating tiles, and return.
September-December	Prototype 3 development, OpenMV introduction, and rescue-kit deployment mechanism development.	Victim detection and rescue-kit deployment.
January-March	Robot (Yatagarasu) development and tire development.	Able to handle obstacles.
April-June	Coordinate management, backup systems, tire improvements, and addition of a 3 x 4 second-floor course.	Coordinate identification, handling various obstacles, improved traversability, and second-floor coordinate setup.

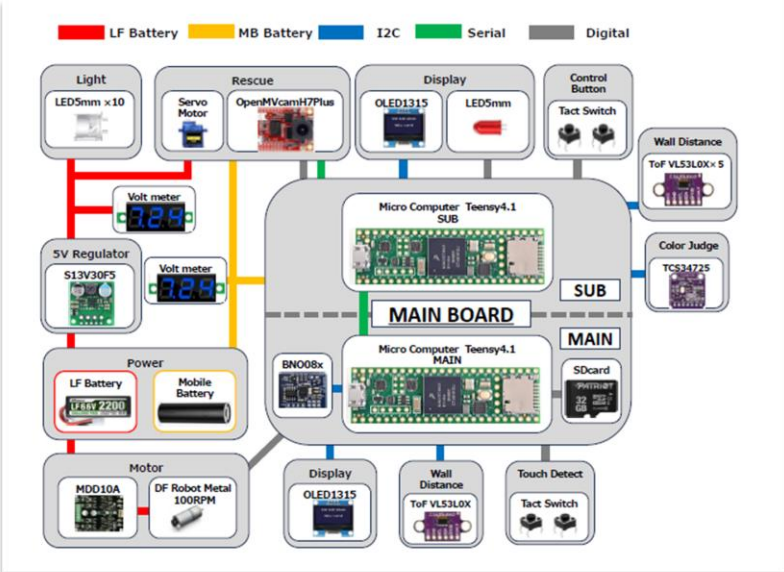
b. Integration Plan

The three team members divided the work and then integrated the results: a robot capable of reliable exploration and return in a flat maze, hardware and programs for obstacles, and a camera program for detecting victims.



Issues Considered

	Person in Charge	Issue	Solution
①	Harada	Reduce errors as much as possible as the highest priority.	The other members designed their work to align with item 1.
②	Sobue	To improve accuracy, we wanted to mount the camera directly sideways on the robot.	We solved this by controlling the robot so that it becomes directly beside the victim without changing the camera position.
③	Tsujimoto	When tires and programs that can traverse bumps and stairs were created, errors also occurred on the flat course.	We did not adopt any development that caused errors on the flat course. We solved this by gradually developing hardware and programs for bumps and stairs while confirming that no errors occurred on the flat course.



Block Diagram

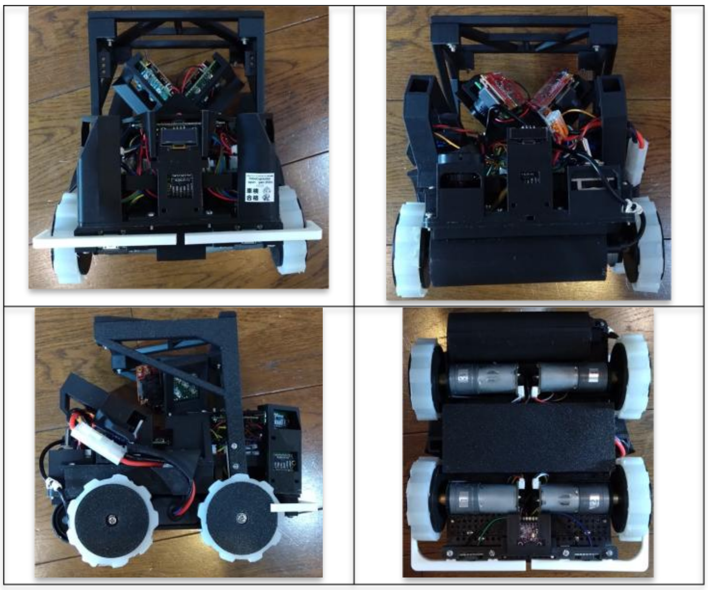
3. Hardware

a. Mechanical Design and Manufacturing

The basic body structure was designed and assembled from scratch using 3D printing. The final specifications are 198 mm wide, 210 mm long, 207 mm high, and approximately 1,550 g in weight.

The body consists of two plates. Heavy batteries and gear motors are mounted under the lower plate to achieve the lowest possible center of gravity. Two motor drivers are mounted on top of the lower plate, and a main board made with a universal PCB is mounted on the upper plate.

The drive system is four-wheel drive using four RB-Dfr-759 gear motors (6 V, 100 RPM). We made our own tires with a diameter of 68 mm, a width of 15 mm, 9 grooves, and a groove depth of 3.0 mm, achieving high traversability.



Size	198 mm (W) x 210 mm (L) x 207 mm (H)
Weight	1,550g
Drive	4WD
Motor	RB-Dfr-759×4
Motor Driver	MDD10A ×2
Tires	Self-made silicone tires (68 mm x 15 mm, 9 grooves, 3.0 mm depth)
Battery	LF battery 6.6 V 2200 mAh
	Mobile battery 5 V 5000 mAh

The rescue kit deployment mechanism is mounted on both sides of the robot. It was designed to align with the camera detection position, allowing rescue kits to be deployed near the victim. The SG90 servo arm only needs to rotate about 100 degrees to deploy a kit. Because the mechanism is very simple, it is easy to maintain and achieved a high success rate in testing.

Rescue Kit Deployment

The tire molds were created with a 3D printer, and liquid silicone was poured into them.

At the Japan Open, we emphasized traversability too much and accuracy became worse, so we reduced the tire size and redeveloped the tires for the World Championship. After various tests, we selected Plan 4 based on the analysis.

	Diameter (mm)	Width (mm)	Number of Grooves	Groove Depth (mm)	Analysis Results		Evaluation
					Traversability	Movement Accuracy	
Plan 1	90	15	8	5	◎	×	×
Plan 2	68	15	0	0	×	△	×
Plan 3	68	15	12	1.5	○-	◎	◎-
Plan 4	68	15	9	3.0	◎+	○+	◎+
Plan 5	68	15	8	3.0	◎−	○-	◎-

b. Innovative Solutions

Improved Part	Details
Low Center of Gravity	All heavy motors and batteries are mounted under the lower plate to maximize the low center of gravity.
Compact Shape	The width is under 200 mm.
Optimized Tire Shape	We tested five designs and selected the optimal shape.
Short I2C Wiring for ToF Sensors	The microcontroller is mounted near the sensors, and the wiring is shortened as much as possible.
Main Board Integrated into One PCB	The main microcontroller, sub microcontroller, and gyro sensor are integrated on one board.
Camera Position and Uniform Lighting	The camera is placed at the upper center to monitor a wide field of view. Lighting is placed in a suitable position so that it clearly illuminates the target without interfering with the camera.
Reliable Left and Right Touch Sensors	We independently developed touch sensors using tact switches that reliably activate when the front touches an object.
Color Sensor Cover	We designed a cover that blocks external light so that stable color values can be read.
Separate Batteries	We use separate batteries for the motors and the control system so that stable control is possible without being affected by motor operation.

c. Electronic Design and Manufacturing

In the electronic design, we focused on stable operation even with many I2C devices. The robot has six VL53L0X ToF distance sensors, one TCS34725 color sensor, and a BNO08x IMU. By minimizing the distance between each sensor and the board, we shortened the wiring and improved stability. We use two OpenMV H7 Plus cameras with built-in microcontrollers. After each camera detects, calculates, and judges a target, it sends the result by serial communication to the sub-control Teensy. Two

MDD10A motor drivers are used to control four motors independently with high precision. The power system combines an LF battery and a mobile battery. This reduces voltage fluctuations in the Teensy and cameras by using a stable 5 V supply that is not affected by ground bounce caused by motor control.

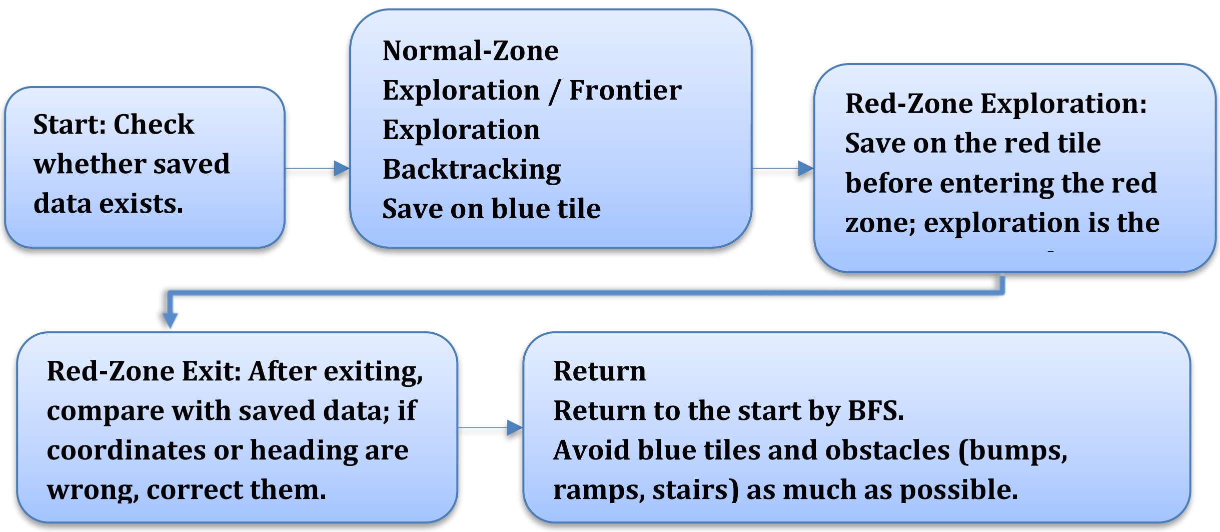
Category	Component	Function
Main/Sub Control	Teensy 4.1×2	The main controller mainly handles the driving system, while the sub controller controls and communicates with sensors, AI cameras, and other peripherals.
Sensors	VL53L0X×6 ※	Measures front, rear, left, right, and diagonal directions. The four main directions determine wall presence, and the diagonal sensors detect walls early.
	BNO08x ※	Used for yaw correction and relative 90-degree turns.
	TCS34725 ※	Used for floor color detection such as black, blue, and red.
	Tact Switch×2	Detects front contact. It consists of two self-made touch bars and two tact switches.
Camera	OpenMV H7 Plus×2	Used to detect letter victims and cognitive targets.
Display	OLED×2 ※	Displays coordinates, sensor values, and other information.
Motor Driver	MDD10A×2	Supports high current and stably controls the four-wheel drive.
Power	LF Battery	For gear motors. 6.6 V, 2200 mAh, stable and high-capacity power source.
	Lithium-ion Battery	5000 mAh, 5 V, for microcontrollers and cameras.
Buttons	Tact Switch×3	Mainly used during development. During competition, used to erase saved data.

※ I2C communication

4. Software

a. Overall Software Architecture

The software is divided into driving control, sensor monitoring, coordinate management, color judgment, victim detection, rescue-kit deployment, and return decisions. Each loop performs periodic processing equivalent to tickall(), so IMU updates, UART reception, and OLED display are not stopped. Heavy processes such as data saving and ToF/color reading are limited to when they are necessary, keeping the control lightweight.



During exploration, each tile is managed as a coordinate (X, Y), and the coordinate is updated when the robot passes 20 cm during a 30 cm move. Wall information is stored for N/E/S/W. Unknown directions are treated as open candidates during exploration; if the robot actually passes through, the direction is updated as no wall, and if both touch sensors hit, it is updated as a wall. For action selection, directions with fewer visits among open candidates are prioritized; ties are resolved in the order left, front, right, and back. During return, the robot uses XY_Number with the start set to 0 and moves to adjacent tiles with smaller numbers.

b. Innovative Solutions

We built several backup systems.

Backup Function	Mechanism
Identify coordinate deviation and return to the correct coordinate.	A microSD card is mounted on the Teensy. When the robot reaches a blue or red tile, it saves the coordinates to the SD card. When leaving the red zone, it reads the coordinate data for the red tile and subsequent tiles. If the coordinates have shifted, the robot moves three tiles and compares the result with the saved data to identify its coordinate.
Color-Tile Detection	Color tiles are checked at 7 cm and 13 cm during travel. After black is detected, the robot moves an additional 2 cm and checks again.
Periodic Wall Attack	To position the robot at the center of each tile, every 30 seconds, if a wall is ahead, the robot intentionally contacts the wall to correct its position and gyro heading.
Time Limits for Each Action	Time limits are set for actions such as robot movement (forward, reverse, and turning), color detection, and coordinate identification. If the time limit is exceeded, the robot automatically selects another action to avoid getting stuck.
Wall-Presence Judgment	Walls are detected by ToF sensors, but if detection is not possible, the state is set to "unknown." If the robot advances and collides, the direction is judged as wall present; if it can pass, it is judged as no wall.

Software Improvements

Software Feature	Details
Floor color is judged by color similarity.	Color similarity is calculated using the weighted Euclidean distance between the measured C/R/G/B values and the initial reference values. The weight reduces the influence of the C value by half to reduce the effect of lighting brightness.
Coordinate Localization	By comparing with saved coordinate data and driving several tiles after a restart, the robot can identify its coordinates and continue running from the saved data. In the red zone, coordinate deviation is likely, so when leaving the red zone, the robot identifies and corrects its coordinates using saved data.
Red Zone Deferred	To ensure reliable saved data, the robot first drives through the simpler area and challenges the red zone later. The distance from the red tile is stored for each coordinate; when the set time arrives, the robot moves to the red tile by the shortest path and enters the red zone.
Reliable Return	For returning, the robot records the distance from the start for each coordinate. When the set time arrives, it returns to the start by the shortest path. Even if the coordinates have shifted, the robot stops once; after restart, it drives several tiles, identifies its coordinates, and can return.
Victim Detection	Cognitive targets are detected using LAB color-space thresholding, while letter victims are identified using machine learning.
Frontier Exploration	The robot stores unexplored coordinates and, if the current position is fully explored, moves toward the nearest unexplored coordinate.
Multi-Floor Coordinate Management	Coordinates can be recorded up to the third floor, and the same X/Y coordinate can be managed as a different floor.

5. Performance Evaluation

Performance was evaluated through step-by-step driving tests and component-level tests. In V1, we confirmed that the robot could drive; in V2, that it could create a map; and in V3, that it could perform rescue actions. In particular, we focused on the return success rate in flat mazes, AI camera detection rate, obstacle traversal by the tires, and color-tile processing.

Evaluation Item	Results / Current Status	Impact on Development
Flat-Maze Return	Reached about 95% by May when no obstacles were present.	Confirmed that the robot can return with high probability in flat mazes.
AI Camera	Cognitive targets: about 50%; letter-victim machine learning: 60% or higher, set as targets and progress values.	Increase OpenMV training data and improve detection conditions.
Tires / Obstacles	Grooveless tires were disadvantageous, and the design with 9 grooves and 3.0 mm depth improved performance on bumps and stairs. Many coordinate deviations occurred on stairs.	Prevent coordinate deviation through software.
Rescue Kit	The simple left/right deployment mechanism achieved a high success rate.	Confirmed 100 consecutive successes in 100 deployments.
Color Tiles	Implemented individual processing for black, blue, and red. Stabilized detection using blue lock and suppression of readings immediately after entering a red tile.	Confirmed 99 successes in 100 tests.
Driving Control	Obstacle traversal performance still needs improvement. On a 36-tile course, return took 6 minutes, so the driving speed needs to be increased.	Handle obstacles through software. Increase Speed from 120 to 140.
Return Accuracy on Obstacle Course	Return success rate is about 60%. If coordinates shift in the normal zone, the robot sometimes cannot return.	Strengthen countermeasures for obstacles placed in the normal zone and prevent coordinate deviation in the normal zone.

Problems and Solutions During Performance Evaluation

Period	Problem	Solution
2025.10	Robot movement was unstable.	Changed the motor driver to MDD10A.
2026.1	The Teensy momentarily lost power.	Thickened the wiring, changed the battery from eneloop pro to an LF battery, added capacitors, and added a second Teensy so that separate main and sub controllers could be used.
2026.4	The robot sometimes ran out of control.	Changed color-tile reading to be led by the main controller.
2026.5	Wall presence was sometimes misjudged.	Any direction that the robot has already passed through is confirmed as no wall.

6. Conclusion

In the development of TOKAI Susano, we first focused on reliable exploration and return in a flat maze without obstacles through three prototypes. After that, toward the World Championship, we gradually improved traversability, coordinate management, and backup systems, developing the robot into its current form. In particular, coordinate recovery using data saved on the SD card, coordinate matching after leaving the red zone, and return control using wall information and exploration history are important technical achievements of this team. Going forward, we will improve the detection success rate for cognitive targets and letter victims, reduce coordinate deviation when driving over obstacles, and improve stability during long runs, aiming for a robot that can steadily accumulate points at the World Championship.